

BREASTFEEDING

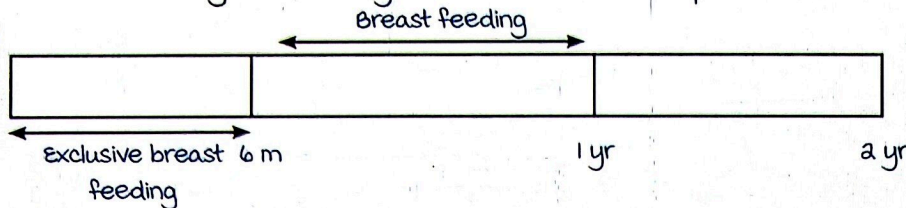
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Introduction

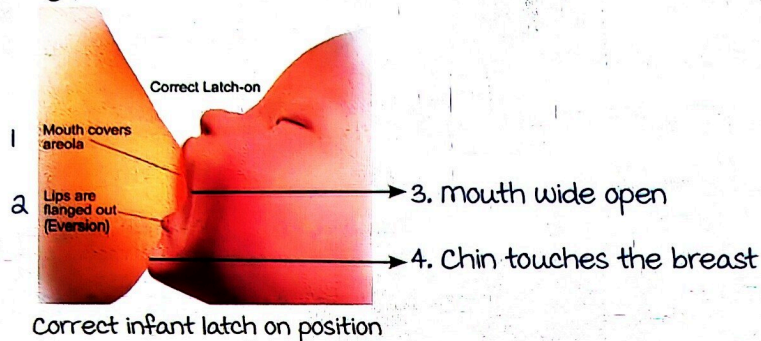
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Initiation of breastfeeding : As early as possible within (minimum : 1 hr after birth).

Exclusive breastfeeding : Infant only receives breast milk upto 6 months of age.



Correct latching position :



Note : upper areola may be visible.

Neonatal reflexes assisting breastfeeding :

1. **Rooting reflex :**

- Nipple touches the mouth & initiates the reflex.

2. **Sucking and swallowing reflex.**

Stimulus (Touch)
around the mouth



Rooting reflex

Feeding in Preterm Babies

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Gestational age at birth	method of feeding
< 28 weeks (↓↓↓ gut motility)	Total Parenteral Nutrition (TPN)
28-31 weeks	Expressed breastmilk via nasogastric or orogastric tube (D/t ↑ risk of aspiration)
32-34 weeks	Feeding breastmilk by paladai or katori spoon (D/t ↑ risk of aspiration)
> 34 weeks	Direct breastfeeding

khanirfan0392@gmail.com

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Nutrition

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Properties of Breast Milk

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1. Immunological properties :mnemonic : **PLABB**.

- Low levels of **Para Amino Benzoic Acid** : Protection from malaria.
- **Lactoferrin** : Protection from *E. coli*.
- **IgA** : **Predominant immunoglobulin** in colostrum.
- **Bifidus factor** : Promotes *Lactobacillus bifidus* growth.
- **Bile salt stimulated lipase** : Protection from giardiasis.
- **Epidermal Growth Factor (EGF)** → Promotes intestinal maturation → ↓ risk of bacterial infection.

2. Human milk Oligosaccharides (HMO) :

- **most abundant solid** in breast milk.
- Non-digestible substance prebiotic → ↑ es favourable gut bacteria → ↓ es risk of diarrhea & **necrotizing enterocolitis**.

3. Nutritional properties :

- Nutritive value : **67 Kcal/100 mL**.

Nutrient	Features
Carbohydrates (Breast milk > cow milk)	• 6-7 g/dL (more than in cow milk). • Lactose → Glucose + Galactose • Galactocerebroside : Promotes myelination.
Proteins (Cow milk > breast milk by 3 times)	• 0.9 - 1.1 g/dL. • ↓ protein → ↓ solute load on kidneys → Kidney protective. • Whey : Casein ratio is 80:20 → Easily digestible. • Amino acids : Taurine & cysteine → Promote brain growth.
Fats (Cow milk = breast milk)	• 3.5 - 4 g/dL. • Rich in PUFA (Eg : DHA : Docosa Hexanoic Acid) : Promotes myelination.

Note :

Cow milk rich in casein → Coagulates in intestine → Curd → Not easily digestible.

Composition of Breast Milk

00:18:13

khanirfan0392@gmail.com

TYPES :**1. Colostrum :**

- Secreted during **first 3-4 days**.
- Thick & lemon yellow-coloured.
- Rich in : (mnemonic → **3As**).
 - **Ig A**.
 - **Vitamin A**.
 - **Anti-infective protein (Lactoferrin)**.

2. Transitional milk :

- Secreted **after 5-14 days.**
- $\uparrow$ sugar, $\uparrow$ fat.
- $\downarrow$ Ig, $\downarrow$ proteins. } Compared to colostrum

3. mature milk :

- Secreted **after 14 days.**

PRETERM MILK :

milk secreted by mother of pre-term baby.

Contains more : (mnemonic : **SIPS**).

- Sodium.
- Iron.
- Protein.
- Sugars (Calories).

FOREMILK AND HINDMILK :

	Foremilk	Hindmilk
Production	At the initiation of feeding	Towards the end of feeding
Consistency	Thin	Thick
main component	Water	Fats
Function	Quenches thirst	Satiety (Energy dense)
Appearance	 White	 Yellow

Deficiencies in Breast Milk

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MICRONUTRIENTS :**1. Vitamins :**

- Vitamin K.
- Vitamin B12 (In vegetarian/vegan mothers)
- Vitamin D.

Supplementation :

Vitamin	Dose
Vitamin K	1 mg IM on anterolateral aspect of thigh (0.5 mg dose given : If birth weight < 1 kg)
Vitamin D	400 IU/ day orally upto 1 yr of age

2. Others :

- Feedback** Iron } Not supplemented d/t good bioavailability
- Zinc }

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SUPPLEMENTATION IN PRETERM BREASTFED BABIES :

Birth weight	Supplements given
1.5-2.5 kg	- Iron (2 mg/kg/day) upto 1 yr of age. - D/t ↓ iron stores & ↓ bioavailability in preterm babies. - Vit D. - Vit K.
< 1.5 kg (Very low birth weight babies)	Preterm birth Human milk fortification (HMF) Vit D, K & iron 32 wks 40 wks

Contraindications of Breast Feeding

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Absolute :

- Congenital lactose intolerance.
- Galactosemia.
- mother on chemo/ radiotherapy.

Relative :

- HIV.
 - SES { High → Formula feeds.
 - Low → Breastfeeding.
 - mixed feeding is contraindicated.
 - mother should be on anti-retroviral therapy.
 - Nevirapine prophylaxis for neonate.
- Tuberculosis :
 - If mother is untreated/Rx < 2 weeks.
- Herpes (Only if active lesions are present on breast).
- Infections.

Expressed Breast Milk Storage

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Temperature	Duration
Room temperature	6-8 hrs
Refrigerator	24 hrs
In a freezer (-20°C)	3 months